

Design and Implementation of an FPGA-Based Bank Queue Management System

Vineeth Mahadevu
Electronics and Communication
Amrita Vishwa Vidyapeetham
Kollam, India
anishchejarla@gmail.com

Namaneni Vivekananda
Electronics and Communication
Amrita Vishwa Vidyapeetham
Kollam, India
email address or ORCID

Modi Ravi Teja
Electronics and Communication
Amrita Vishwa Vidyapeetham
Kollam, India
mchanukya2439@gmail.com

Murapaka Venkatesh
Electronics and Communication
Amrita Vishwa Vidyapeetham
Kollam, India
msdsvenkatesh008@gmail.com

Kambham Mourya
Electronics and Communication
Amrita Vishwa Vidyapeetham
Kollam, India
mourya80008@gmail.com

Abstract—In today's fast-paced banking environments, efficient customer flow management is essential for enhancing service quality and reducing waiting times. A real-time Bank Queue Management System has been designed and implemented using FPGA technology to enhance customer service efficiency in banking environments. Utilizing the parallel processing capabilities of the Artix-7 FPGA on the Basys 3 development board, the system is modeled in Verilog HDL and incorporates key features such as token-based customer management, passcode-protected administrative control, and real-time status display through on-board peripherals like switches, push buttons, LEDs, and seven-segment displays. The architecture employs finite state machines (FSMs) and debouncing techniques to ensure reliable, low-latency operation. Simulation and hardware verification using Xilinx Vivado confirmed accurate performance under different operating scenarios, making this hardware-based solution highly suitable for real-time queue management in banks, hospitals, and other customer service centers.

Index Terms—FPGA, Queue Management System, Finite State Machine (FSM), Xilinx Vivado, Basys 3.

I. LITERATURE SURVEY

Queue management systems play a vital role in streamlining customer flow and enhancing service efficiency in various sectors such as banking, healthcare, and retail. In traditional bank service environments, long waiting times and unmanaged customer queues often lead to customer dissatisfaction and operational delays. To address these challenges, electronic queue management solutions have been introduced, offering automated token distribution, real-time queue tracking, and dynamic service counter allocation. The primary objective of such systems is to ensure orderly customer service while minimizing wait times and improving overall user experience. In the context of this project, the focus is on developing a real-time, hardware-implemented queue management system specifically tailored for banking environments using FPGA technology, enabling faster response times and reliable control logic compared to conventional software-based systems.

Finite State Machines (FSMs) are widely used in hardware design for implementing real-time control systems, especially when sequential logic and predictable state transitions are critical. In their study, Sundararajan and Malarvizhi [1] presented an FPGA-based Traffic Light Controller implemented using FSM methodology on the Xilinx Spartan-3E board. Their design efficiently handled different traffic flow scenarios by controlling traffic lights based on state transitions derived from traffic conditions. The project showcased how FSMs can be effectively modeled in Verilog HDL for real-time hardware control applications. This paper serves as a relevant reference for understanding FSM-driven control flow design on FPGA platforms, which directly relates to our implementation of customer token control and teller state transitions in the Bank Queue Management System.

Another important aspect in real-time hardware systems is ensuring reliable user input handling. Mechanical switches and buttons, such as those used for customer token requests and teller control in our project, are prone to noise and signal bouncing. Ganssle [2], in his widely referenced work "A Guide to Debouncing," provides an in-depth comparative study of both software and hardware debouncing techniques. His paper discusses the limitations of software debouncing in time-critical embedded systems and emphasizes the advantages of hardware-based debouncing methods, especially when implemented on FPGAs. Drawing from these insights, our system incorporates a dedicated hardware debouncer module that ensures clean, single-event detection for each button press, preventing multiple unwanted triggers during customer or teller interactions.

Together, these studies highlight the importance of both state-driven control logic using FSMs and robust hardware input handling through debouncing in developing reliable real-time service automation systems like the one presented in this paper.

II. METHODOLOGY

The project begins with identifying the key functional requirements such as passcode-based administrative access, token generation, teller control and real-time status display. A modular architecture is adopted, integrating finite state machines (FSMs) for system control and hardware debouncing techniques for reliable input handling. Each module is designed, simulated, and verified before deployment on the Basys 3 FPGA board to ensure real-time and error-free operation in a banking service environment.

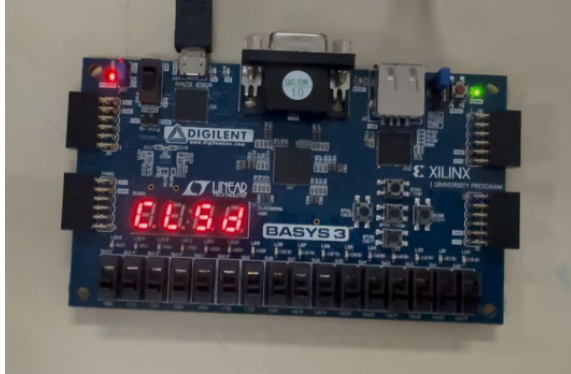


Fig. 1. SSD showing bank is closed

Fig.1 illustrates the system state when the bank is closed, which is indicated by the seven-segment display (SSD) showing the message "CLSD." This condition occurs when the reset button (BTND) is pressed, triggering the system to transition to the Bank Closed state through the finite state machine (FSM) logic. In this state, the token distribution is halted, teller operations are disabled, and the SSD provides a clear visual indication to users that the bank is not currently in service.

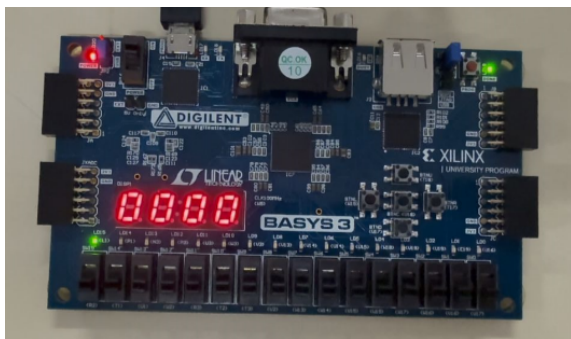


Fig. 2. LED showing bank is open

Fig.2 depicts the condition where the bank is open, indicated by the leftmost LED (LD15) turning ON when the corresponding bank open/close control switch (SW[15]) is set to the ON state. This action triggers a state transition in the system's FSM to the Bank Open state, enabling customer token requests and teller operations. The LED acts as a visual indicator for

both customers and bank staff, confirming that the bank is operational and ready to serve customers.

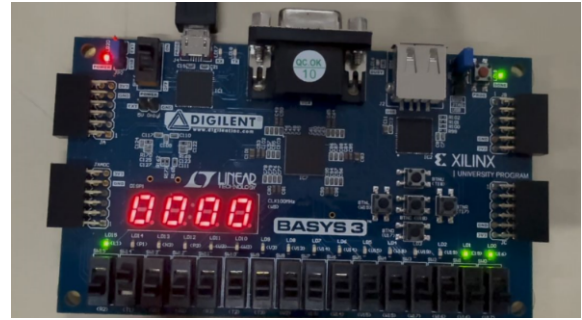


Fig. 3. LED's showing number of counters opened

Fig.3 shows the condition where counters are opened based on the state of switch SW[14] and the corresponding push button inputs. When SW[14] is set to ON, the system enables teller control, allowing counters to be opened when the respective teller push buttons (BTNL for Counter 1 and BTNR for Counter 2) are pressed. Once a teller presses their button, the system's FSM transitions that counter to the open state, and the status is visually indicated by turning ON the corresponding LEDs (LD0 for Counter 1 and LD1 for Counter 2). This ensures that counters open only when both SW[14] is ON and the teller initiates the action by pressing the button, providing controlled and flexible counter management.

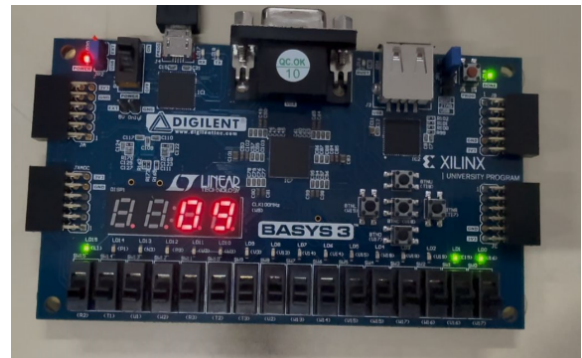


Fig. 4. SSD showing number of tokens distributed

Fig.4 illustrates the condition where the seven-segment display (SSD) shows the number of tokens distributed as customers press the UP push button (BTNU). Each button press triggers the system's FSM to generate a new token number, incrementing the count sequentially. The updated token number is displayed in real-time on the SSD, allowing both customers and staff to monitor the token distribution process. Additionally, the SSD is programmed to briefly blink the new token number when issued, providing clear visual feedback that the token has been successfully generated. This process helps manage customer flow efficiently and ensures that each customer receives a unique service token.

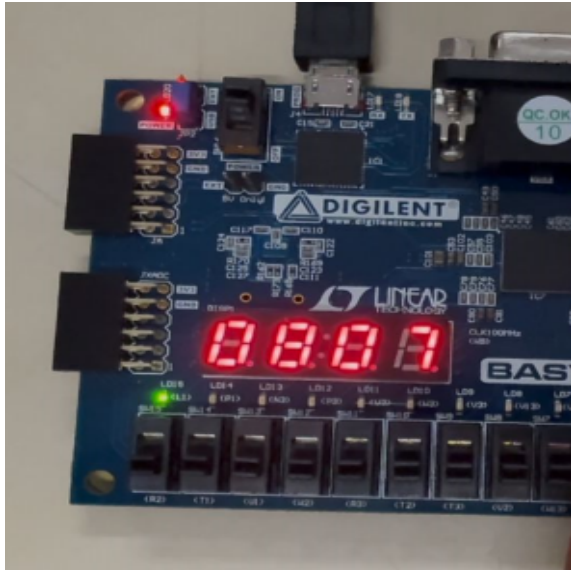
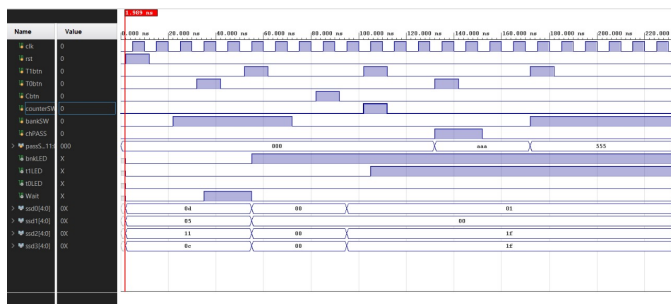


Fig. 5. SSD showing token call at the counters

Fig.5 illustrates the condition where the seven-segment display (SSD) shows the token number being called at the respective counters. When a teller presses their designated counter button (BTNL for Counter 1 or BTNR for Counter 2), the system's FSM updates the SSD to display the next token number in the queue, indicating to the customer that it is their turn to proceed to the counter. This real-time token call display helps maintain an orderly service flow and ensures clear communication between the service counters and waiting customers.

RESULTS

The simulation of the FPGA-Based Bank Queue Management System was performed using Xilinx Vivado, focusing on verifying state transitions and output responses. The timing diagrams clearly show the system's behavior during token generation, counter activation, and display updates. The results confirm that the FSM logic and debouncing modules work as expected, ensuring accurate and noise-free output for each input condition.



FUTURE WORK

The current implementation efficiently manages two counters with real-time control and display using onboard peripherals. As a future enhancement, the system can be expanded to support more counters by introducing multi-click detection on existing teller buttons. For example, double-clicking BTNL could represent Counter 3, while double-clicking BTNR could represent Counter 4, allowing the existing hardware inputs to handle additional service points without the need for extra buttons. Another improvement could involve enabling individual teller passcode configuration, allowing each teller to set and update their own password for accessing or controlling their assigned counters, thus enhancing security and flexibility. Additionally, considering the limited display capacity of the current seven-segment display (SSD), an external LCD module could be integrated to display larger token numbers, detailed system messages, and support for additional counters, providing better visibility and user communication. These enhancements would improve system scalability and user interaction while maintaining the real-time performance benefits of FPGA-based design.

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